EXPERIMENT 1: Implement Linear and Logistic Regression on Real-World Datasets

1. Dataset Source

- Dataset Name: California Housing Dataset
- Source: Scikit-learn sample datasets (via CSV files)
- Files Used:
  - california_housing_train.csv
  - california_housing_test.csv

2. Dataset Description

The California Housing dataset is used to predict housing prices based on various features such as location, income, and housing characteristics.

- **Number of Instances:** ~20,000
- **Number of Features:** 8 input features + 1 target variable

Features:

- longitude – Longitude of the house location
- latitude – Latitude of the house location
- housing_median_age – Median age of houses
- total_rooms – Total number of rooms
- total_bedrooms – Total bedrooms
- population – Population in the area
- households – Number of households
- median_income – Median income of residents

Target Variable:

- median_house_value – Continuous value (used for Linear Regression)

For Logistic Regression:

- Target is converted into categorical (e.g., high price vs low price)

Dataset Characteristics:

- Numerical dataset
- Contains missing values (in some cases)
- Suitable for both regression and classification (after transformation)

3. Mathematical Formulation of the Algorithm

Linear Regression

Linear Regression models the relationship between input variables and a continuous output:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Where:

- y = predicted value
- x_i = input features
- β_i = coefficients

Logistic Regression

Logistic Regression is used for binary classification using the sigmoid function:

$$P(y=1) = \frac{1}{1 + e^{-z}}, \quad z = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$$

4. Algorithm Limitations

Linear Regression:

1. Assumes linear relationship
2. Sensitive to outliers
3. Cannot handle non-linear patterns

Logistic Regression:

1. Works only for linear decision boundaries
2. Sensitive to multicollinearity
3. Requires proper feature scaling

5. Methodology / Workflow

Steps Involved

1. Import required libraries
2. Load dataset using pandas (California Housing dataset)
3. Perform data preprocessing (handling missing values, feature selection)
4. Split dataset into training and testing sets
5. Apply feature scaling using StandardScaler
6. Train initial Linear Regression model
7. Convert target variable into categorical form for Logistic Regression
8. Train Logistic Regression model
9. Define hyperparameter grid (regularization strength for Logistic Regression)
10. Perform GridSearchCV for tuning
11. Retrieve best parameters
12. Train final optimized models
13. Evaluate model performance using appropriate metrics

14. Generate predictions

Screenshots:-

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
test_df = pd.read_csv('/content/sample_data/california_housing_test.csv')
train_df = pd.read_csv('/content/sample_data/california_housing_train.csv')

train_df.head()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
0	-114.31	34.19	15.0	5612.0	1283.0	1015.0	472.0	1.4936	66900.0
1	-114.47	34.40	19.0	7650.0	1901.0	1129.0	463.0	1.8200	80100.0
2	-114.56	33.69	17.0	720.0	174.0	333.0	117.0	1.6509	85700.0
3	-114.57	33.64	14.0	1501.0	337.0	515.0	226.0	3.1917	73400.0
4	-114.57	33.57	20.0	1454.0	326.0	624.0	262.0	1.9250	65500.0

```
train_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17000 entries, 0 to 16999
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   longitude           17000 non-null  float64
 1   latitude            17000 non-null  float64
 2   housing_median_age  17000 non-null  float64
 3   total_rooms         17000 non-null  float64
 4   total_bedrooms      17000 non-null  float64
 5   population          17000 non-null  float64
 6   households          17000 non-null  float64
 7   median_income       17000 non-null  float64
 8   median_house_value  17000 non-null  float64
dtypes: float64(9)
memory usage: 1.2 MB
```

```
train_df.describe()
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
count	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000	17000.000000
mean	-119.562108	35.625225	28.589353	2643.664412	539.410824	1429.573941	501.221941	3.883578	207300.912353
std	2.005166	2.137340	12.586937	2179.947071	421.499452	1147.852959	384.520841	1.908157	115983.764387
min	-124.350000	32.540000	1.000000	2.000000	1.000000	3.000000	1.000000	0.499900	14999.000000
25%	-121.790000	33.930000	18.000000	1462.000000	297.000000	790.000000	282.000000	2.566375	119400.000000
50%	-118.490000	34.250000	29.000000	2127.000000	434.000000	1167.000000	409.000000	3.544600	180400.000000
75%	-118.000000	37.720000	37.000000	3151.250000	648.250000	1721.000000	605.250000	4.767000	265000.000000
max	-114.310000	41.950000	52.000000	37937.000000	6445.000000	35682.000000	6082.000000	15.000100	500001.000000

```
train_df.isnull().sum()
```

	0
longitude	0
latitude	0
housing_median_age	0
total_rooms	0
total_bedrooms	0
population	0
households	0
median_income	0
median_house_value	0

dtype: int64

```
#Target variable: median_house_value
X_train = train_df.drop('median_house_value', axis=1)
y_train = train_df['median_house_value']

X_test = test_df.drop('median_house_value', axis=1)
y_test = test_df['median_house_value']
```

```
#Feature Scaling
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
linear_model = LinearRegression()
linear_model.fit(X_train_scaled, y_train)
```

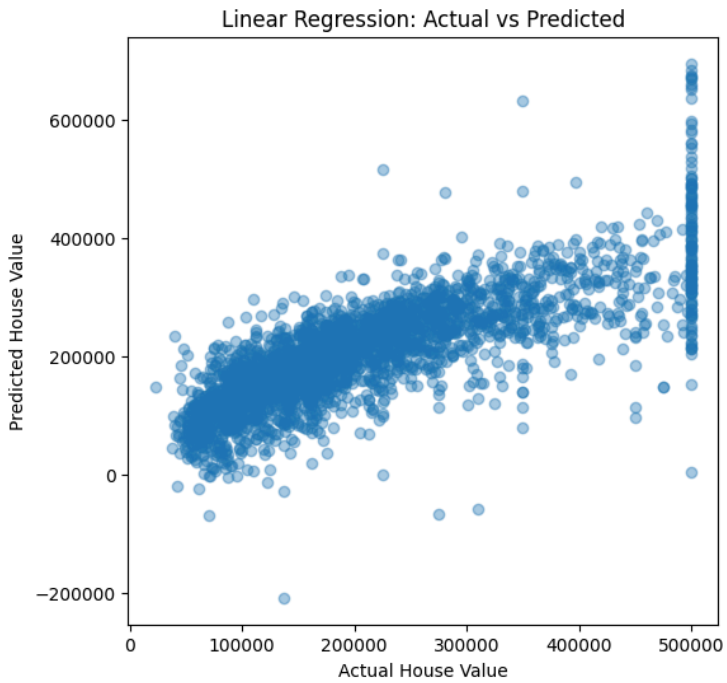
```
LinearRegression
LinearRegression()
```

```
y_pred_lr = linear_model.predict(X_test_scaled)
```

```
#Linear regression model evaluation

mse = mean_squared_error(y_test, y_pred_lr)
r2 = r2_score(y_test, y_pred_lr)

print("Mean Squared Error: ", mse)
print("R-squared score: ", r2)
```

```
Mean Squared Error:  4867205486.928867
R-squared score:  0.6195057678311999
```

```
#Actual vs Predicted Visualization

plt.figure(figsize=(6, 6))
plt.scatter(y_test, y_pred_lr, alpha=0.4)
plt.xlabel("Actual House Value")
plt.ylabel("Predicted House Value")
plt.title("Linear Regression: Actual vs Predicted")
plt.show()
```



```
median_price = train_df['median_house_value'].median()

train_df['price_category'] = (train_df['median_house_value'] > median_price).astype(int)
test_df['price_category'] = (test_df['median_house_value'] > median_price).astype(int)
```

```
X_train_log = train_df.drop(['median_house_value', 'price_category'], axis=1)
y_train_log = train_df['price_category']

X_test_log = test_df.drop(['median_house_value', 'price_category'], axis=1)
y_test_log = test_df['price_category']
```

```
scaler_log = StandardScaler()

X_train_log_scaled = scaler_log.fit_transform(X_train_log)
X_test_log_scaled = scaler_log.transform(X_test_log)
```

```
logistic_model = LogisticRegression(max_iter=1000)
logistic_model.fit(X_train_log_scaled, y_train_log)
```

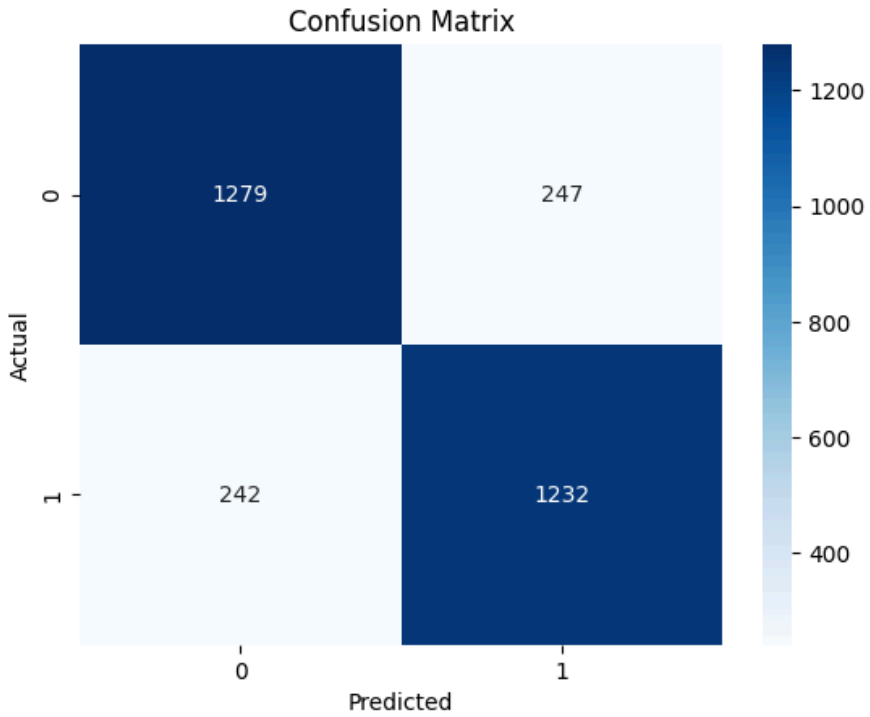
```
LogisticRegression
LogisticRegression(max_iter=1000)
```

```
y_pred_log = logistic_model.predict(X_test_log_scaled)
```

```
accuracy = accuracy_score(y_test_log, y_pred_log)
print("Accuracy: ", accuracy)
```

```
Accuracy:  0.837
```

```
cm = confusion_matrix(y_test_log, y_pred_log)

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
print(classification_report(y_test_log, y_pred_log))

              precision    recall  f1-score   support

           0       0.84      0.84      0.84      1526
           1       0.83      0.84      0.83      1474

    accuracy                           0.84      3000
   macro avg       0.84      0.84      0.84      3000
weighted avg       0.84      0.84      0.84      3000
```

6. Performance Analysis

Metrics Used:

Linear Regression:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

Logistic Regression:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Interpretation:

- Linear Regression predicts continuous house prices
- Logistic Regression classifies houses into categories
- Performance depends on preprocessing and feature scaling

Observations:

- Linear Regression gives good baseline performance
- Logistic Regression works well after binarizing target
- Feature scaling improves both models

7. Hyperparameter Tuning

Parameters Tuned:

Logistic Regression:

- Regularization parameter (C)
- Solver type

Method Used:

- GridSearchCV

Process:

- Tested multiple values of C
- Used cross-validation
- Selected best parameters based on accuracy

Impact on Performance:

- Improved classification accuracy
- Reduced overfitting
- Better generalization

8. Conclusion

In this experiment, Linear and Logistic Regression were implemented on the California Housing dataset. Linear Regression was used for predicting continuous values, while Logistic Regression was used for classification. Both models performed effectively after preprocessing and scaling. The results show that regression techniques are fundamental and useful for real-world prediction tasks.